

# JEE MAIN REVISION MASTER: HALOALKANES AND HALOARENES

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 177 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                  | Frequency | Category        |
|---------------|-----------------------------------------------|-----------|-----------------|
| Very High     | Haloalkanes And Haloarenes - Core Application | 177 PYQs  | General Concept |

## 2. Master Formula Reference Card

| Concept / Topic | Mathematical Formulation & Governing Equations |
|-----------------|------------------------------------------------|
|-----------------|------------------------------------------------|

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Haloalkanes And Haloarenes - Core Application (Very High Yield — 177 PYQs)

**Core Principle:** Key problem pattern in Haloalkanes And Haloarenes.

■■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

**Representative Source PYQs:**

- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : Due to increase in van der Waals forces, the order of boiling points is  $CH_3CH_2CH_3 > \dots$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Given below are two statements : Statement (I) : Benzyl chloride reacts faster in  $S_N1$  mechanism than ethyl chloride. Statement (II) :  $E_{\text{act}}$ ..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Consider the following compounds Arrange these compounds in the increasing order of reactivity with nitrating mixture...." [Ans: N/A]

## 4. Benchmark Illustrative PYQs with Master Solutions

**Benchmark Problem 1: JEE Main 2026 (Online) 21st January Evening Shift (Q#1)**

The correct order of the rate of the reaction for the following reaction with respect to nucleophiles is :  $\text{CH}_3\text{Br} + \text{Nu}^- \rightarrow \text{CH}_3\text{Nu} + \text{Br}^-$

- (A)  $\text{ClO}_4^- > \text{CH}_3\text{COO}^- > \text{OH}^- > \text{PO}_4^{3-}$
- (B)  $\text{OH}^- > \text{PO}_4^{3-} > \text{CH}_3\text{COO}^- > \text{ClO}_4^-$  **[Correct]**
- (C)  $\text{CH}_3\text{COO}^- > \text{PO}_4^{3-} > \text{OH}^- > \text{ClO}_4^-$
- (D)  $\text{PO}_4^{3-} > \text{OH}^- > \text{CH}_3\text{COO}^- > \text{ClO}_4^-$

**Correct Answer: B**

**Solution Analysis:** For  $\text{CH}_3\text{Br}$  (a methyl halide), the reaction proceeds by  $\text{S}_{\text{N}}2$ , so the rate increases with nucleophilicity of  $\text{Nu}^-$ . For oxygen nucleophiles in the same row, nucleophilicity roughly follows basicity, and any resonance stabilization of the anion reduces nucleophilicity:  $\text{OH}^-$  : strong base, charge localized  $\Rightarrow$  best nucleophile  
 $\text{PO}_4^{3-}$  : resonance-stabilized  $\Rightarrow$  less nucleophilic than  $\text{OH}^-$   
 $\text{CH}_3\text{COO}^-$  : even more resonance delocalization over two oxygens  $\Rightarrow$  weaker nucleophile  
 $\text{ClO}_4^-$  : highly resonance-stabilized, essentially non-nucl